

Mid Semester Examination 2025

Name of the Program: BTech(ECE)
 Name of the Course: OOPs Using C++

Semester: VI
 Course Code: TOE-610

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Explain the concept of object-oriented programming paradigm. What is the significance of private, protected and public specifiers in a class? Explain with the help of example.	.(CO1,CO2)
OR		
(b)	Explain the difference between static function and instance function, static data member and instance data member?	
Q2	(10 marks)	
(a)	Explain following with help of C++ program. (i) Passing objects as call by value in a member function. (ii) Passing objects as call by reference in a member function.	(CO1,CO2)
OR		
(b)	Explain following with help of C++ program. (i) Scope Resolution Operator : : (ii) Any five string class functions.	
Q3	(10 marks)	
(a)	Write a C++ program to input a string and convert lowercase alphabet into uppercase alphabet and vice versa. Example: EXAMiNaTion Output: examINAtION	(CO1,CO2)
OR		
(b)	Write a C++ program to demonstrate the concept of Compile time polymorphism.	
Q4	(10 marks)	
(a)	Define a class named DataChange with following description: Data Members private int x , y Member function setValue(int,int)→ this member function will store value in x and y. Member function showData()→ this member function will print value of x and y. Create a member function to swap data member value of two different object of the same class .	(CO1,CO2)
OR		
(b)	What is "this" pointer? Give an example to illustrate the use of it in C++.	
Q5	(10 marks)	
(a)	Explain constructor and destructor with proper C++ program..	(CO1,CO2)
OR		
(b)	Write a C++ program to create a class Called NDemo with following members: Data Member: int tnum Parametrized constructor to store value into tnum	

Function Members:	
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	void checkValue()→ This member function will check whether first and last digit of a given number num are equal or not..
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	In main function create object of class NDemo and perform the above task.
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